

Лабораторная работа № 4

Базовая настройка HTTP-сервера Apache

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Цель:

Получить навыки установки и базового конфигурирования Apache HTTP Server.



Установка Apache HTTP Server

```
root@server:~
PS D:\work\ndbaranov\vagrant> ssh vagrant server
ssh: Could not resolve hostname vagrant: ??? ???? ??????????
PS D:\work\ndbaranov\vagrant> vagrant ssh server
vagrant : The term 'vagrant' is not recognized as the name of a cmdlet, function, script file, or opera
ble program. Check the spelling of the name, or if a path was included, verify that the path is correct
and try again.
At line:1 char:1
+ vagrant ssh server
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (vagrant:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS D:\work\ndbaranov\vagrant> vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please enter
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sat Sep  5 20:29:54 2026 from 10.0.2.2
vagrant@server:~$ sudo -i
root@server:~# dnf -y groupinstall "Basic Web Server"
Last metadata expiration check: 2:01:44 ago on Sat 05 Sep 2026 06:57:05 PM UTC.
Dependencies resolved.
=====
Package                Architecture      Version           Repository        Size
=====
Installing group/module packages:
httpd                  x86_64            2.4.63-13.el10_2.6  appstream         47 k
httpd-manual            noarch            2.4.63-13.el10_2.6  appstream        2.2 M
mod_fcgid               x86_64            2.3.9-35.el10      appstream         72 k
mod_ssl                 x86_64            1:2.4.63-13.el10_2.6 appstream        110 k
Installing dependencies:
apr                     x86_64            1.7.5-3.el10       appstream         127 k
apr-util               x86_64            1.6.3-23.el10_1    appstream         97 k
apr-util-ldb           x86_64            1.6.3-23.el10_1    appstream         13 k
httpd-core              x86_64            2.4.63-13.el10_2.6  appstream        1.4 M
httpd-filesystem        noarch            2.4.63-13.el10_2.6  appstream         14 k
httpd-tools             x86_64            2.4.63-13.el10_2.6  appstream         81 k
rocky-logos-httpd       noarch            100.5-3.el10       appstream         24 k
Installing weak dependencies:
apr-util-openssl        x86_64            1.6.3-23.el10_1    appstream         15 k
mod_http2               x86_64            2.0.29-4.el10_2.2  appstream        164 k
mod_lua                 x86_64            2.4.63-13.el10_2.6  appstream         59 k
Installing Groups:
Basic Web Server

Transaction Summary
=====
Install 14 Packages

Total download size: 4.4 M
Installed size: 14 M
Downloading Packages:
(1/14): apr-1.7.5-3.el10.x86_64.rpm                243 kB/s | 127 kB  00:00
(2/14): apr-util-ldb-1.6.3-23.el10_1.x86_64.rpm     25 kB/s | 13 kB   00:00
```

Смысл результата

На server установлен пакет httpd вместе с зависимостями. После установки проверено наличие исполняемых файлов и системного unit-файла службы.

Запуск и автозапуск httpd

```
root@server:~  
root@server:~# firewall-cmd --add-service=http  
success  
root@server:~# firewall-cmd --add-service=http --permanent  
success  
root@server:~# systemctl enable httpd  
systemctl start httpd  
systemctl status httpd --no-pager  
Created symlink '/etc/systemd/system/multi-user.target.wants/httpd.service' → '/usr/lib/systemd/system/httpd.service'.  
● httpd.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)  
   Active: active (running) since Sat 2026-09-05 21:01:24 UTC; 42ms ago  
 Invocation: c72533a2046c4d4e9fd2740555d735b5  
    Docs: man:httpd.service(8)  
 Main PID: 13682 (httpd)  
   Status: "Started, listening on: port 443, port 80"  
    Tasks: 178 (limit: 10234)  
  Memory: 17.2M (peak: 17.3M)  
     CPU: 169ms  
   CGroup: /system.slice/httpd.service  
           └─13682 /usr/sbin/httpd -DFOREGROUND  
             └─13683 /usr/sbin/httpd -DFOREGROUND  
               └─13684 /usr/sbin/httpd -DFOREGROUND  
                 └─13685 /usr/sbin/httpd -DFOREGROUND  
                   └─13686 /usr/sbin/httpd -DFOREGROUND  
                     └─13687 /usr/sbin/httpd -DFOREGROUND  
  
Sep 05 21:01:24 server.ndbaranov.net systemd[1]: Starting httpd.service - The Apache HTTP Server...  
Sep 05 21:01:24 server.ndbaranov.net (httpd)[13682]: httpd.service: Referenced but unset environme...TIONS  
Sep 05 21:01:24 server.ndbaranov.net systemd[1]: Started httpd.service - The Apache HTTP Server.  
Sep 05 21:01:24 server.ndbaranov.net httpd[13682]: Server configured, listening on: port 443, port 80  
Hint: Some lines were ellipsized, use -l to show in full.  
root@server:~#
```

Смысл результата

Apache запущен через systemctl и включён в автозагрузку. Статус active (running) подтверждает успешную загрузку конфигурации.

Открытие HTTP в firewalld

```
root@client:~  
root@client:~# curl http://192.168.1.1  
<!doctype html>  
<html>  
<head>  
<meta charset='utf-8'>  
<meta name='viewport' content='width=device-width, initial-scale=1'>  
<title>HTTP Server Test Page powered by: Rocky Linux</title>  
<style type='text/css'>  
/**/<br/><br/>html {<br/>  height: 100%;<br/>  width: 100%;<br/>}<br/>  body {<br/>background: rgb(69,23,32);<br/>background: -moz-linear-gradient(180deg, rgba(69,23,32,1) 30%, rgba(0,0,0,1) 90%) ;<br/>background: -webkit-linear-gradient(180deg, rgba(69,23,32,1) 30%, rgba(0,0,0,1) 90%) ;<br/>background: linear-gradient(180deg, rgba(69,23,32,1) 30%, rgba(0,0,0,1) 90%);<br/>background-repeat: no-repeat;<br/>background-attachment: fixed;<br/>filter: progid:DXImageTransform.Microsoft.gradient(startColorstr="#b43c56",endColorstr="#b43c56",GradientType=1);<br/>color: white;<br/>font-size: 0.9em;<br/>font-weight: 400;<br/>font-family: 'Montserrat', sans-serif;<br/>margin: 0;<br/>padding: 10em 6em 10em 6em;<br/>box-sizing: border-box;<br/><br/>  }<br/><br/>h1 {<br/>  text-align: center;<br/>  margin: 0;<br/>  padding: 0.6em 2em 0.4em;<br/>  color: #fff;<br/>  font-weight: bold;<br/>  font-family: 'Montserrat', sans-serif;<br/>  font-size: 2em;<br/>}<br/>h1 strong {<br/>  font-weight: bolder;<br/>  font-family: 'Montserrat', sans-serif;<br/>}<br/>h2 {<br/>  font-size: 1.5em;<br/>  font-weight: bold;<br/>}<br/><br/>.title {<br/>  border: 1px solid black;<br/>  font-weight: bold;</pre></div><div data-bbox="680 184 829 215" data-label="Section-Header"><h2>Смысл результата</h2></div><div data-bbox="680 241 925 369" data-label="Text"><p>В постоянную конфигурацию межсетевого экрана добавлен сервис http. После reload порт 80 доступен клиентскому узлу.</p></div><div data-bbox="961 956 986 976" data-label="Page-Footer"><p>5/9</p></div>
```

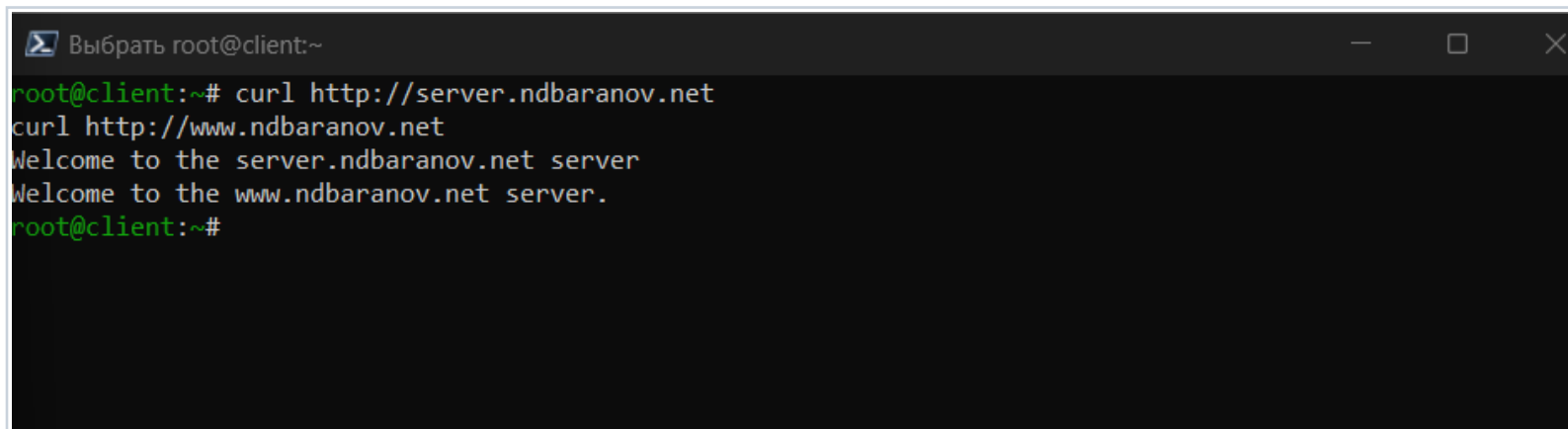
Веб-контент и имена узлов

```
root@server:~
GNU nano 8.1 /var/named/master/fz/ndbaranov.net
$ORIGIN .
$TTL 86400      ; 1 day
ndbaranov.net   IN SOA  ndbaranov.net. server.ndbaranov.net. (
                        2026090503 ; serial
                        86400      ; refresh (1 day)
                        3600       ; retry (1 hour)
                        604800     ; expire (1 week)
                        10800      ; minimum (3 hours)
                        )
                NS    ndbaranov.net.
                A     192.168.1.1
$ORIGIN ndbaranov.net.
$TTL 2400      ; 40 minutes
client         A     192.168.1.30
               DHCID  ( AAEBnDYKK/CtPCj+e5omkyL8RyndF4Fc95e2r6pNeTxW
                       PvA= ) ; 1 1 32
$TTL 86400     ; 1 day
dhcp           A     192.168.1.1
ns             A     192.168.1.1
server         A     192.168.1.1
www            A     192.168.1.1
```

Смысл результата

Созданы страницы для server.ndbaranov.net и www.ndbaranov.net, а DNS-зона дополнена соответствующими А-записями. Имена обоих сайтов указывают на один сервер.

Проверка сайтов через curl

A terminal window with a dark background and light green text. The window title is "Выбрать root@client:~". The terminal shows the following commands and output:

```
root@client:~# curl http://server.ndbaranov.net
curl http://www.ndbaranov.net
Welcome to the server.ndbaranov.net server
Welcome to the www.ndbaranov.net server.
root@client:~#
```

Смысл результата

С клиента выполнены HTTP-запросы к обоим доменным именам. В ответ получено содержимое созданных страниц, следовательно DNS, сеть и Apache работают совместно.

Provision-сценарий Apache

```
PowerShell 7 (x64)
PS D:\work\ndbaranov\vagrant> vagrant validate
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
Vagrantfile validated successfully.
PS D:\work\ndbaranov\vagrant> vagrant provision server --provision-with "server http"
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Running provisioner: server http (shell)...
server: Running: C:/Users/barash/AppData/Local/Temp/vagrant-shell20260906-27448-1uwwwn.sh
server: Provisioning HTTP server
server: Last metadata expiration check: 2:21:41 ago on Sat 05 Sep 2026 06:57:05 PM UTC.
server: Dependencies resolved.
server: =====
server: Package           Architecture Version           Repository        Size
server: =====
server: Installing Groups:
server: Basic Web Server
server:
server: Transaction Summary
server: =====
server: Complete!
server: Warning: ALREADY_ENABLED: 'http' already in 'public'
server: success
server: Warning: ALREADY_ENABLED: http
server: success
server: HTTP provisioning completed
PS D:\work\ndbaranov\vagrant>
```

Смысл результата

Сценарий server-http автоматизирует установку пакета, открытие порта и создание контента. Повторный запуск завершён без ошибок.

Результат

Apache запущен и доступен по HTTP; страницы двух доменных имён открываются с клиентского узла. DNS, firewalld и веб-сервер работают как единая цепочка.